HR Data Preparation Evidence Package

Monthly workforce dataset for the HR business review

Prepared by	[Your name]
Date prepared	2026-10-28
Role and scenario	HR analyst at a fictional mid-sized company preparing the monthly workforce dataset for an HR business review
Tools used	Microsoft Excel, Microsoft Power Query, Excel PivotTable (Data Model)
Source files	Monthly_Workforce_Extract.csv; HR_Reference_Extract.csv; HR_Project_Validation_Rules.xlsx
Final output	Final_Analysis_Ready_Workforce.csv: one flat table, 1 header row, 10 records, 11 fields (10 required + ValidationStatus)
Copilot	Copilot was not used for this project (Section 8).
Data	Synthetic project files only. No real workforce information is included.

Results at a glance

Monthly rows in	Final records out	Headcount	Total FTE	Excluded	Duplicate removed	Unmatched (not added)	Supported corrections
12	10	10	9.5	1 (E1008)	1 (E1011)	1 (E1012)	6 values in 5 records

The final dataset was produced by one refreshable Power Query workflow. Raw source files were never edited, and no cells were changed by hand. Every correction, exclusion and enrichment traces to a rule in `HR_Project_Validation_Rules.xlsx`. Every corrected value also traces to a matching EmployeeID in `HR_Reference_Extract.csv`.

A value was corrected only when the supplied evidence supported it. One record (E1008) failed the date-logic rule. Neither source contains evidence to fix its dates, so it was excluded and logged for follow-up rather than guessed.

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1. Source Review

Before building the dataset, I established what each source represents, how the records connect, which rules apply, and what evidence exists to support any change.

1.1 Source inventory (Governance Record §1)

Source file	What the source represents	Reporting period / effective date	Source status	Notes
Monthly_Workforce_Extract.csv	Primary monthly workforce extract. This is the starting reporting population.	No period field in the file. Latest date in the file is 2026-07-18.	Supplied raw extract: 12 rows, 9 fields. Contains data-quality issues. Read-only; never edited.	Source of the population (VR-07) and of all output fields except Region.
HR_Reference_Extract.csv	HR reference attributes per employee: department, region, worker status, worker type.	EffectiveDate 2026-08-01 (all 12 rows).	Supplied: 12 rows, 6 fields, UTF-8 with BOM. Reference evidence only.	Authoritative Department names (VR-04) and Region (VR-10). Never used to add records.
HR_Project_Validation_Rules.xlsx	Source of truth: 11 validation rules, Approved Values, Metric Reference, Output Specification.	Not applicable.	Supplied. VR-05, VR-07 and the population rule are marked “Confirm – provisional pending SME validation”.	Approved Values sheet is loaded directly into Power Query (`Src_ApprovedValues`).
Sample_Analysis_Ready_Workforce.csv	Example of the expected flat-table CSV structure.	Not applicable.	Structure reference only.	Only the column layout was followed. No values or decisions were used.
Completed_Example_HR_Evidence_Package.docx	Example of how an evidence package can be organised.	Not applicable.	Separate example; not an answer key.	Only the layout was used, not the content.

1.2 How the records connect

`EmployeeID` is the only key shared by both sources. Once the exact duplicate is removed, the relationship is one-to-one. Field alignment:

Monthly extract field	Reference field	Relationship	How it is used
EmployeeID	EmployeeID	Join key	Left-outer merge from the monthly extract (population) to the reference (evidence).
Department	DepartmentName	Same attribute, different name	Reference name is authoritative where an ID matches (VR-04).
EmploymentStatus	WorkerStatus	Same attribute	Corroborates status standardization (VR-02).
EmploymentType	WorkerType	Same attribute	Corroborates type standardization (VR-03).
Location	—	Monthly only	Kept from the monthly extract.
—	Region	Reference only	Added to the output for matched IDs (VR-10).
EmployeeName, HireDate, TerminationDate, FTE	—	Monthly only	No reference evidence exists for these fields, so they cannot be corrected from the reference.
—	EffectiveDate	Reference only	Shows how current the reference is (2026-08-01). Not output.

1.3 Applicable rules and the reporting population

- **Population (VR-07, Output Specification):** start with the EmployeeIDs in the monthly extract. Remove only confirmed exact duplicates (VR-06) and records a rule requires to be excluded (VR-05). Never add reference-only IDs.
- **Field-level rules:** VR-01 EmployeeID required; VR-02 status; VR-03 employment type; VR-04 department; VR-08 FTE range; VR-10 Region from reference.
- **Record-level rules:** VR-05 TerminationDate on or after HireDate; VR-06 exact duplicate rows.
- **Output rules:** VR-09 consistent names and labels, YYYY-MM-DD dates, one flat table; VR-11 minimum required fields.
- **Reporting definitions (Metric Reference):** Headcount = distinct EmployeeID in the final dataset; FTE = sum of FTE in the final dataset.

2. Reconciliation

The sources were reconciled on `EmployeeID` with the `Recon_IDs` query (full outer join of monthly IDs against reference IDs). Field values were compared with the `Recon_Field_Compare` query, which compares raw monthly values with the reference.

2.1 Record counts

Check	Monthly extract	HR reference	Result
Rows	12	12	Equal row counts, but they hide a duplicate and an unmatched ID.
Distinct EmployeeIDs	11	12	Monthly has one repeated ID.
Repeated EmployeeIDs	1 (E1011 × 2 rows)	0	E1011 rows compared field by field (see 2.4).
Blank EmployeeIDs	0	0	VR-01 passes.
Matched EmployeeIDs	11	11	Every monthly ID has a reference record.
Monthly-only IDs	0	—	No monthly record lacks reference evidence.
Reference-only IDs	—	1 (E1012)	Documented as unmatched; not added (VR-07).

2.2 Record-count bridge: source to final output

Step	Records	Change	Rule
Raw monthly rows	12	—	—
Remove exact duplicate row (second E1011 row)	11	−1	VR-06
Exclude date-logic failure (E1008)	10	−1	VR-05
Reference-only IDs (E1012)	10	0 (not added)	VR-07
Final analysis-ready dataset = CSV rows = PivotTable headcount	**10**		**VR-11**

2.3 EmployeeID match results (Recon_IDs)

EmployeeID	Monthly rows	Match status	Repeated in monthly	Final disposition
E1001	1	Matched	No	Retained
E1002	1	Matched	No	Retained (status standardized)
E1003	1	Matched	No	Retained

EmployeeID	Monthly rows	Match status	Repeated in monthly	Final disposition
E1004	1	Matched	No	Retained (department and type corrected)
E1005	1	Matched	No	Retained
E1006	1	Matched	No	Retained
E1007	1	Matched	No	Retained (blank department filled from reference)
E1008	1	Matched	No	**Excluded: VR-05 date-logic failure**
E1009	1	Matched	No	Retained (department corrected)
E1010	1	Matched	No	Retained (status standardized)
E1011	2	Matched	**Yes**	**One row retained; exact duplicate removed (VR-06)**
E1012	0	Reference only	No	**Not added; documented as unmatched (VR-07)**

2.4 Material field differences between matched records (Recon_Field_Compare)

Across the 11 matched IDs, the raw Department agrees with the reference for 8 records, EmploymentStatus for 9 (exact text) and EmploymentType for 10. The six differences are listed below. After the supported standardizations, all 11 records agree with the reference on all three fields. Monthly Location is consistent with reference Region for every record: each city maps to exactly one region, so there are no conflicts.

EmployeeID	Field	Monthly value	Reference value	Assessment	Treatment
E1002	EmploymentStatus	ACTIVE	Active	Case variant of an approved value	Standardize (VR-02)
E1004	Department	IT	Information Technology	Abbreviation; not an approved name	Correct from reference (VR-04)
E1004	EmploymentType	Full Time	Full-Time	Missing hyphen; not an approved value	Standardize (VR-03)
E1007	Department	(blank)	Customer Support	Required value missing	Fill from reference (VR-04)
E1009	Department	Ops	Operations	Abbreviation; not an approved name	Correct from reference (VR-04)
E1010	EmploymentStatus	“active ” (lower case, trailing space)	Active	Case and spacing variant of an approved value	Trim and standardize (VR-02)

2.5 Visible difference or confirmed error?

Observation	Classification	Evidence	Treatment
E1011 appears twice	Confirmed error: exact duplicate	All 9 fields identical. The reference holds one E1011 record. Nothing supports multiple assignments.	One row retained (VR-06)
E1008 TerminationDate 2023-05-15 is before HireDate 2023-06-01	Confirmed rule failure (VR-05)	The dates contradict each other. The reference agrees on status (Terminated) but holds no dates that could correct them.	Excluded; follow-up (Section 5)

Observation	Classification	Evidence	Treatment
Labels IT, Ops, blank department, ACTIVE, “active ”, Full Time	Confirmed non-approved values with supporting evidence	Approved Values sheet plus the matching reference record for each ID	Corrected (Section 4.1)
E1003 is Terminated (2026-07-18)	Visible difference only; not an error	TerminationDate is after HireDate. The reference also shows Terminated. No rule removes terminated employees.	Retained unchanged
E1009 is a Contractor at FTE 1.0	Visible difference only; not an error	FTE is within the VR-08 range. The reference also shows Contractor.	Retained unchanged
E1012 is in the reference only	Unmatched record; not a monthly-data error	VR-07: the reference is evidence only	Documented; not added

3. Power Query Transformations

All preparation logic lives in Power Query, so the workflow can be repeated by refreshing the queries. The raw CSV files are read as-is, and every change is a named Applied Step. A single `SourceFolder` parameter holds the file location, and **Data > Refresh All** re-runs the whole pipeline from source to final table.

3.1 Query inventory

Query	Load setting	Purpose
SourceFolder	Parameter	Folder path for all source files (the only place the path is stored).
Src_Monthly	Connection only	Imports the raw monthly extract unchanged (all text).
Src_Reference	Connection only	Imports the HR reference. Trims text, types EffectiveDate, and prefixes fields with `Ref_`.
Src_ApprovedValues	Connection only	Loads the Approved Values sheet from the rules workbook so approved lists are never hard-coded.
Recon_IDs	Table, Reconciliation sheet	Full outer join of monthly and reference IDs; row counts per ID; match status.
Recon_Field_Compare	Table, Reconciliation sheet	Raw monthly vs reference values for Department, Status and Type.
Workforce_Validated	Table, Validation sheet	Main query: removes duplicates, cleanses, enriches from the reference, and validates. Holds all 11 unique records with ValidationStatus and ExceptionReason.
Workforce_Exceptions	Table, Exceptions sheet	Records whose ValidationStatus is not Valid.
Workforce_Final	Table, Final sheet	Final reporting population. Source of the CSV export and the PivotTable.

Dependency chain: Src_Monthly + Src_Reference + Src_ApprovedValues → Workforce_Validated → Workforce_Final → CSV export and PivotTable (Workforce_Exceptions also reads Workforce_Validated).

3.2 Workforce_Validated: Applied Steps

INSERT SCREENSHOT — Capture A: Power Query Editor: Workforce_Validated query with the Applied Steps pane

Show: the Queries pane (left) with all query names, the query name `Workforce_Validated`, and the full **Applied Steps** pane (right) listing the 19 steps below.

If the data preview makes the image unreadable, capture the left and right panes as two focused images.

Replace this box with the image (Insert > Pictures, about 17 cm / 6.7 in wide), then delete this box. Keep the caption below.

Capture A. Power Query Editor: Workforce_Validated query with the Applied Steps pane

#	Applied step	What it does	Rule
1	Source	References `Src_Monthly` (raw extract, 12 rows).	—
2	Removed Exact Duplicates	`Table.Distinct` on raw rows. Runs before any cleansing, so only true exact duplicates collapse (E1011). Result: 11 rows.	VR-06
3	Trimmed Text	Removes leading and trailing spaces from every column (fixes “active ”).	VR-09
4	Blanks To Null	Turns empty strings into null so missing values are detectable (E1007 Department, blank TerminationDates).	VR-01, VR-09
5	Changed Type	HireDate and TerminationDate to Date, FTE to Decimal (en-US). No conversion errors.	VR-09
6	Standardized Status	Proper case on EmploymentStatus: ACTIVE and active become Active.	VR-02

#	Applied step	What it does	Rule
7	Standardized Type	Replaces “Full Time” with “Full-Time”.	VR-03
8	Merged HR Reference	Left-outer merge on EmployeeID to `Src_Reference`. The monthly extract stays the population.	VR-07
9	Expanded HR Reference	Brings in Ref_Department and Ref_Region.	VR-04, VR-10
10	Department From Reference	Uses the reference department name wherever a match exists (IT, blank, Ops corrected).	VR-04
11	Region From Reference	Adds Region from the matched reference record.	VR-10
12–14	ApprovedStatus, ApprovedType, ApprovedDepartment	Approved lists read from `Src_ApprovedValues` (rules workbook).	Approved Values
15	Added Exception Reason	Runs every check and lists the failed rule IDs per record.	VR-01, 02, 03, 04, 05, 08, 10
16	Added Validation Status	VR-05 failure gives Excluded; any other failure gives Flagged; otherwise Valid.	VR-05, VR-07
17	Removed Helper Columns	Drops Ref_Department after use.	VR-09
18	Reordered Columns	Puts the columns in the required field order.	VR-11
19	Sorted By EmployeeID	Ascending sort for review.	—

3.3 Workforce_Final: Applied Steps

INSERT SCREENSHOT — Capture B: Power Query Editor: Workforce_Final query with Applied Steps and the 10-row result

Show: the query name `Workforce_Final`, the **Applied Steps** pane (4 steps), and the data preview with EmployeeID visible (10 rows, E1008 absent).

Replace this box with the image (Insert > Pictures, about 17 cm / 6.7 in wide), then delete this box. Keep the caption below.

Capture B. Power Query Editor: Workforce_Final query with Applied Steps and the 10-row result

#	Applied step	What it does	Rule
1	Source	References `Workforce_Validated` (11 records).	—
2	Removed Excluded Records	Keeps records whose ValidationStatus is not Excluded. Removes E1008. Result: 10 rows.	VR-05, VR-07
3	Selected Output Fields	Keeps the 10 required fields plus ValidationStatus, in the required order.	VR-11
4	Formatted Dates YYYY-MM-DD	Converts dates to yyyy-MM-dd text so the CSV format does not depend on Windows locale settings.	VR-09

3.4 Repeatability controls

- The raw source files are never opened for editing. All changes are Applied Steps that re-run on refresh.
- No manual cell edits were made to any loaded table. The CSV was exported from the refreshed `Workforce_Final` table.
- Approved values come from the rules workbook, not typed lists, so a rule change flows through on refresh.

- Corrections are rule-driven (reference match or approved-value mapping), not keyed to specific EmployeeIDs, so they also work on next month's extract.

4. Cleansing and Validation

4.1 Cleansing actions

EmployeeID	Field	Before	After	Supporting evidence	Applied step
E1002	EmploymentStatus	ACTIVE	Active	Approved Values; reference WorkerStatus = Active (VR-02)	Standardized Status
E1010	EmploymentStatus	“active ”	Active	Approved Values; reference WorkerStatus = Active (VR-02)	Trimmed Text, Standardized Status
E1004	EmploymentType	Full Time	Full-Time	Approved Values; reference WorkerType = Full-Time (VR-03)	Standardized Type
E1004	Department	IT	Information Technology	Reference DepartmentName for E1004 (VR-04)	Department From Reference
E1007	Department	(blank)	Customer Support	Reference DepartmentName for E1007 (VR-04)	Department From Reference
E1009	Department	Ops	Operations	Reference DepartmentName for E1009 (VR-04)	Department From Reference
All 11	Region	(not in monthly extract)	From reference	Reference Region for each matched ID (VR-10)	Region From Reference
E1011	Whole row	2 identical rows	1 row	All 9 fields identical (VR-06)	Removed Exact Duplicates
All	HireDate, TerminationDate	Text	Date, output as YYYY-MM-DD	VR-09 date format	Changed Type, Formatted Dates YYYY-MM-DD

4.2 Values deliberately not changed

- **E1008 HireDate and TerminationDate:** no supplied source holds alternative dates. Swapping or guessing them would invent data, so the record is excluded instead (Section 5).
- **E1003 (Terminated):** valid dates, and the reference agrees. No rule removes terminated employees from the population.
- **EmployeeName, Location, FTE:** all values pass their checks. The reference holds no competing values, so they are kept exactly as supplied.

4.3 Field-level and record-level validation results

Rule	Level	Check	Tested	Pass	Fail	Outcome
VR-01	Field	EmployeeID is not blank	12 raw rows	12	0	Pass
VR-02	Field	EmploymentStatus is Active or Terminated	11	11	0	Pass after 2 supported standardizations
VR-03	Field	EmploymentType is Full-Time, Part-Time or Contractor	11	11	0	Pass after 1 supported standardization
VR-04	Field	Department is an approved name; reference name used where matched	11	11	0	Pass after 3 reference corrections

Rule	Level	Check	Tested	Pass	Fail	Outcome
Type	Field	HireDate, TerminationDate convert to Date; FTE converts to Decimal; HireDate present	11	11	0	Pass; no conversion errors
VR-05	Record	TerminationDate on or after HireDate (2 records have a TerminationDate)	11	10	1	**E1008 fails: Excluded**
VR-06	Record	No exact duplicate rows; one row per EmployeeID	12 raw rows	11	1 duplicate	E1011 duplicate removed; 11 unique IDs
VR-07	Population	Monthly IDs matched to reference; no reference-only additions	11 monthly IDs	11 matched	—	E1012 (reference only) documented, not added
VR-08	Field	FTE > 0 and ≤ 1.0	11	11	0	Pass (values 0.5–1.0)
VR-10	Field	Region populated from the reference match	11	11	0	Pass; none flagged
VR-09	Output	Consistent names and labels, YYYY-MM-DD dates, one flat table	Final table	Pass	—	Pass
VR-11	Output	All 10 required fields present; optional field fully populated	Final table	Pass	—	Pass: 10 required + ValidationStatus

4.4 Validation output (Workforce_Validated, all 11 unique records)

INSERT SCREENSHOT — Capture C: Validation sheet: Workforce_Validated output with ValidationStatus and ExceptionReason

Show: the loaded `Workforce_Validated` table with the **EmployeeID**, **ValidationStatus** and **ExceptionReason** columns visible, including the E1008 row marked Excluded.

Replace this box with the image (Insert > Pictures, about 17 cm / 6.7 in wide), then delete this box. Keep the caption below.

Capture C. Validation sheet: Workforce_Validated output with ValidationStatus and ExceptionReason

Employee ID	Department	Region	Status	Type	HireDate	Termination Date	FTE	Validation Status	Exception Reason
E1001	Human Resources	Central	Active	Full-Time	2022-03-14		1.0	Valid	
E1002	Finance	East	Active	Full-Time	2021-07-05		1.0	Valid	
E1003	Sales	South	Terminated	Full-Time	2020-11-02	2026-07-18	1.0	Valid	
E1004	Information Technology	West	Active	Full-Time	2023-01-09		1.0	Valid	
E1005	Sales	South	Active	Part-Time	2024-05-20		0.5	Valid	
E1006	Finance	East	Active	Full-Time	2022-09-12		1.0	Valid	
E1007	Customer Support	Central	Active	Full-Time	2025-02-03		1.0	Valid	
E1008	Operations	South	Terminated	Full-Time	2023-06-01	2023-05-15	1.0	**Excluded**	VR-05 TerminationDate before HireDate
E1009	Operations	South	Active	Contractor	2025-08-11		1.0	Valid	
E1010	Marketing	East	Active	Full-Time	2024-10-01		1.0	Valid	
E1011	Marketing	East	Active	Full-Time	2026-01-15		1.0	Valid	

5. Exceptions

5.1 Exception and Disposition Log (Governance Record §3)

Employee or record ID	Issue	Evidence reviewed	Disposition	Rationale
E1008	TerminationDate 2023-05-15 is earlier than HireDate 2023-06-01 (VR-05 failure).	Monthly extract dates; HR reference (status Terminated, Operations, no date fields); VR-05.	**Excluded** from the final dataset; documented for follow-up with the HR source owner or SME.	No supplied evidence shows which date is wrong. Correcting it would be a guess. VR-05 requires exclusion when no evidence supports a correction.
E1011	EmployeeID appears twice; both rows are identical in all 9 fields.	Recon_IDs (Monthly_Rows = 2); field-by-field comparison; single E1011 reference record; VR-06.	**Duplicate removed**; one authoritative record retained.	Exact duplicate confirmed. Nothing in the sources supports multiple assignments.
E1012	In the HR reference only; no monthly record.	Recon_IDs (MatchStatus = Reference only); VR-07; Output Specification.	**Documented as unmatched; not added.**	The population starts with monthly IDs. The reference is evidence, not a population source.
E1007	Department blank.	Reference DepartmentName = Customer Support; VR-04.	**Corrected** from the reference.	An authoritative reference match exists for the same EmployeeID.
E1004	Department “IT” and EmploymentType “Full Time” are not approved values.	Reference: Information Technology / Full-Time; Approved Values; VR-03, VR-04.	**Corrected** to the approved values.	Both corrections are supported by the reference and by the approved list.
E1009	Department “Ops” is not an approved name.	Reference DepartmentName = Operations; VR-04.	**Corrected** from the reference.	Supported by the reference match.
E1002, E1010	EmploymentStatus “ACTIVE” and “active ” (case and trailing space).	Approved Values; reference WorkerStatus = Active for both; VR-02.	**Standardized** to Active.	Formatting variants of an approved value, confirmed by the reference.

Disposition summary: 1 excluded (E1008); 1 duplicate removed (E1011); 1 unmatched reference record not added (E1012); 5 records corrected or standardized; 0 records retained but flagged (no VR-08 or VR-10 failures occurred).

5.2 Follow-up items

Item	Action required	Owner	Status
E1008 dates	Confirm the correct HireDate and TerminationDate with the HR source system. Once corrected at source, a refresh will bring the record back into the dataset automatically.	HR data owner / SME	Open
E1012 unmatched	Confirm why E1012 is in the reference but not in the monthly extract (for example, extract timing).	HR data owner	Open
Provisional rules	VR-05, VR-07 and the population rule are marked “provisional pending SME validation”. Obtain SME sign-off.	HR SME	Open

6. PivotTable Summary

The PivotTable is a reporting-readiness check built from the final validated table. It is not the source of truth for the dataset; the CSV is.

6.1 PivotTable configuration

Data source	Excel table `Workforce_Final` (10 rows), the same table exported to the CSV. Added to the Data Model so a distinct count is available.
Rows	Department
Values	Headcount = Distinct Count of EmployeeID; Total FTE = Sum of FTE (Metric Reference sheet)
Filter	EmploymentStatus = (All)

INSERT SCREENSHOT — Capture D: PivotTable: Headcount and Total FTE by Department, built from Workforce_Final

Show: the PivotTable with the EmploymentStatus filter, the Department rows, Headcount and Total FTE, the Grand Total row, and the **PivotTable Fields** pane showing the fields used.

Check first: PivotTable Analyze > Change Data Source must point to `Workforce_Final`.

Replace this box with the image (Insert > Pictures, about 17 cm / 6.7 in wide), then delete this box. Keep the caption below.

Capture D. PivotTable: Headcount and Total FTE by Department, built from Workforce_Final

6.2 Summary values

Department	Headcount	Total FTE
Customer Support	1	1.0
Finance	2	2.0
Human Resources	1	1.0
Information Technology	1	1.0
Marketing	2	2.0
Operations	1	1.0
Sales	2	1.5
Grand Total	**10**	**9.5**

6.3 Tie-out to the final CSV

- Headcount 10 = 10 data rows in the CSV = 10 distinct EmployeeIDs (no duplicates remain).
- Total FTE 9.5 = sum of the FTE column in the CSV.
- E1008 (excluded) and E1012 (reference only) are absent from both the PivotTable and the CSV.

Note on terminated employees. Following the Metric Reference definition, headcount counts every distinct EmployeeID in the final dataset. That includes E1003, terminated 2026-07-18 and retained because no supplied rule removes terminated employees.

Setting the EmploymentStatus filter to Active gives 9 employees and 8.5 FTE. That is shown for stakeholder context only; the rule-defined figures are 10 and 9.5.

7. Governance and Traceability

This section completes the HR_Project_Governance_Record_Template. Template sections that sit naturally elsewhere in the package are cross-referenced below, so the source-to-output path can be reviewed from a single place.

7.1 Where each governance record section is completed

Governance Record template section	Location in this package
§1 Source Inventory	Section 1.1
§2 Preparation Controls	Section 7.2
§3 Exception and Disposition Log	Section 5.1
§4 Definitions and Assumptions	Section 7.3
§5 Copilot Use Record	Section 8
§6 Final Traceability Check	Section 7.5

7.2 Preparation Controls (Governance Record §2)

Control / check	Purpose	Evidence location	Result / status	Notes
Record-count reconciliation	Every source row is accounted for in the output	Section 2.1, 2.2	Pass: 12 → 11 → 10, each step explained	—
EmployeeID match	Relate monthly records to reference evidence (VR-07)	Section 2.3; Recon_IDs	Pass: 11 matched, 0 monthly-only, 1 reference-only	E1012 documented
Exact-duplicate check	Keep one authoritative record (VR-06)	Section 2.3; step Removed Exact Duplicates	1 duplicate removed (E1011)	Runs on raw rows
Field-difference review	Separate visible differences from confirmed errors	Section 2.4, 2.5; Recon_Field_Compare	6 differences; all supported corrections	—
Approved-value checks	VR-02, VR-03, VR-04	Section 4.3; ExceptionReason column	Pass for 11 of 11 after standardization	Lists read from the rules workbook
Required-field and type checks	VR-01, VR-11; data types	Section 4.3; step Changed Type	Pass	No conversion errors
Date-logic check	VR-05	Section 4.3, 4.4; Capture C	1 failure (E1008), excluded	Follow-up open
FTE range check	VR-08	Section 4.3	Pass (0.5–1.0)	—
Region enrichment	VR-10	Section 4.1; step Region From Reference	11 of 11 populated	None flagged
Output format	VR-09, VR-11	Section 3.3; Capture B	Pass	Dates are yyyy-MM-dd text
PivotTable-to-CSV tie-out	Metric Reference	Section 6.3	Pass: 10 / 9.5 both ways	—
Repeatability	No undocumented manual edits	Section 3.4	Pass: Refresh All rebuilds the output	SourceFolder parameter

7.3 Definitions and Assumptions (Governance Record §4)

Definition / assumption	Approved source	How applied	Notes
Reporting population	VR-07; Output Specification	Monthly EmployeeIDs, minus the confirmed exact duplicate and the VR-05 exclusion. No reference-only IDs.	Provisional pending SME validation
Headcount	Metric Reference	Distinct Count of EmployeeID on Workforce_Final = 10	Includes terminated E1003
FTE	Metric Reference	Sum of FTE on Workforce_Final = 9.5	—
Approved values	Approved Values sheet	Loaded into Power Query via Src_ApprovedValues and used in the validation checks	Not hard-coded
Department authority	VR-04	The reference DepartmentName replaces the monthly value wherever an EmployeeID matches	All 11 matched
Region source	VR-10; Output Specification	Region taken only from the matched reference record	Never inferred from Location
Exact duplicate	VR-06	Rows identical in every raw field, tested before cleansing	Near-duplicates are not collapsed
Status and type variants	VR-02, VR-03	Case, spacing and hyphen variants of an approved value are treated as that value	Assumption; confirmed by the matching reference record for each case
Terminated employees	VR-07; Metric Reference	Retained in the final dataset and in headcount	No supplied rule excludes them
Reporting period	Not stated in the monthly extract	Treated as the same cycle as the reference (EffectiveDate 2026-08-01)	Assumption; no date-based filtering applied
Optional field ValidationStatus	VR-11; Output Specification	Included and populated for every row (all Valid)	Does not replace any required field
Date format	VR-09; Output Specification	YYYY-MM-DD; blank TerminationDate where none	—

7.4 Source-to-output traceability

Output field	Source	Transformation (query: step)	Rule
EmployeeID	Monthly.EmployeeID	Workforce_Validated: Trimmed Text	VR-01
EmployeeName	Monthly.EmployeeName	Workforce_Validated: Trimmed Text	—
Department	Reference.DepartmentName (matched on EmployeeID)	Workforce_Validated: Department From Reference	VR-04
Location	Monthly.Location	Workforce_Validated: Trimmed Text	—
Region	Reference.Region (matched on EmployeeID)	Workforce_Validated: Region From Reference	VR-10
EmploymentStatus	Monthly.EmploymentStatus	Workforce_Validated: Trimmed Text, Standardized Status	VR-02
EmploymentType	Monthly.EmploymentType	Workforce_Validated: Standardized Type	VR-03
HireDate	Monthly.HireDate	Changed Type (Date); Workforce_Final: Formatted Dates YYYY-MM-DD	VR-05, VR-09
TerminationDate	Monthly.TerminationDate	Blanks To Null, Changed Type; Workforce_Final: Formatted Dates YYYY-MM-DD	VR-05, VR-09
FTE	Monthly.FTE	Workforce_Validated: Changed Type (Decimal)	VR-08

Output field	Source	Transformation (query: step)	Rule
ValidationStatus	Derived	Workforce_Validated: Added Exception Reason, Added Validation Status	VR-05, VR-07

Record path: Monthly_Workforce_Extract.csv (12 rows) → Src_Monthly → Workforce_Validated (11 records; 1 Excluded) → Workforce_Final (10 records) → Final_Analysis_Ready_Workforce.csv (10 records) → PivotTable (Headcount 10, FTE 9.5).

7.5 Final Traceability Check (Governance Record §6)

Check	Status	Evidence
The final CSV is derived from the reviewed and validated data state.	Confirmed	The CSV was exported from the refreshed Workforce_Final table (Section 3.3, Capture B).
The PDF evidence package reflects the same final data state.	Confirmed	Sections 2.2, 4.4 and 6 show the same 10 records, headcount 10 and FTE 9.5 as the CSV.
All material exceptions have a documented disposition.	Confirmed	Section 5.1 covers E1008, E1011, E1012 and all corrected records.
Any Copilot-assisted work includes the prompt, relevant output, verification performed, and decision.	Not applicable	Copilot was not used (Section 8).

8. Copilot Use

> **Copilot was not used for this project.** No Copilot prompts or outputs were used in reconciliation, Power Query transformations, cleansing, validation, exception decisions, the PivotTable or this evidence package.

Copilot Use Record (Governance Record §5)

Project step	Prompt	Relevant Copilot output	Verification performed	Decision: Accept / Revise / Reject
Steps 1–5 (all)	None: Copilot not used	None	Not applicable. Results were verified by the query-based checks and the CSV/PivotTable tie-out (Sections 4.3, 6.3).	Not applicable

9. Final Review Against the Success Checklist

HR_Project_Success_Checklist item	Status	Where evidenced
Source and reconciliation		
Only the supplied synthetic project files were used.	✓	Section 1.1; cover page
Source files and their purposes are identifiable.	✓	Section 1.1
Record-count, duplicate, unmatched or conflicting-record checks are documented.	✓	Sections 2.1 to 2.5
Power Query and cleansing		
The transformation workflow is repeatable and understandable.	✓	Sections 3.1, 3.4
The PDF includes readable Power Query transformation evidence.	✓	Captures A and B; Sections 3.2, 3.3
Supported standardization decisions are documented; uncertain values are not silently guessed.	✓	Sections 4.1, 4.2
Validation and exceptions		
Field-level and record-level validation checks are visible.	✓	Section 4.3; Capture C
Meaningful failed checks have documented dispositions.	✓	Section 5.1
The final dataset reflects the reviewed exception decisions.	✓	Sections 2.2, 3.3, 6.3
Reporting and governance		
At least one PivotTable summary is built from the final validated dataset.	✓	Section 6; Capture D
Source traceability, definitions and assumptions, controls, and exception notes are documented.	✓	Sections 7.2 to 7.4, 5.1
Any Copilot-assisted step includes the prompt, output, verification and decision.	✓	Section 8 (Copilot not used)
Final submission cross-check		
The PDF is readable and tells the preparation story from source to final output.	✓	Sections 1 to 7
The CSV contains one flat analysis-ready table with a header row.	✓	Final_Analysis_Ready_Workforce.csv
The CSV matches the final reviewed data state shown in the evidence.	✓	Sections 4.4, 6.3, 7.5
Only the required PDF and CSV are prepared for LMS submission.	✓	This PDF + Final_Analysis_Ready_Workforce.csv

Submitted files: (1) Final_Analysis_Ready_Workforce.csv (one flat table, 1 header row, 10 records); (2) this PDF evidence package.